

HARRINGTON, PE, Canada

WMO: 713500

Lat: 46.3436N

Lon: 63.1697W

Elev: 53

StdP: 100.69

Time Zone: -4.00 (NAA)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-18.7	-16.5	-23.4	0.5	-18.1	-21.1	0.6	-15.6	13.3	-4.9	12.0	-3.8	4.3	280	0.639

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.9	26.7	20.5	25.4	19.6	24.1	18.9	21.8	25.2	20.9	24.0	20.1	22.9	4.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.6	15.3	24.0	19.8	14.6	23.0	18.9	13.8	22.0	64.0	25.3	60.8	24.1	57.8	22.9	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	9.8	8.7	DB	-21.5	29.5	1.9	1.4	-22.9	30.5	-24.0	31.3	-25.0	32.0	-26.4	33.0	(4)
(5)			WB	-21.9	23.2	1.9	0.7	-23.3	23.7	-24.4	24.1	-25.4	24.5	-26.8	25.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.3	-6.0	-6.3	-2.8	3.4	9.1	14.3	19.2	19.0	14.9	9.3	3.6	-2.3	(6)	
	DBStd	9.89	5.76	5.16	4.45	3.93	3.90	3.63	2.69	2.50	3.53	3.82	4.49	5.05	(7)	
	HDD10.0	2269	497	457	398	203	66	6	0	0	3	61	198	382	(8)	
	HDD18.3	4487	756	691	656	449	287	127	23	23	112	282	442	640	(9)	
	CDD10.0	934	0	0	1	4	37	136	284	279	151	37	6	0	(10)	
	CDD18.3	109	0	0	0	0	1	7	48	43	10	1	0	0	(11)	
	CDH23.3	447	0	0	0	0	7	33	211	165	30	1	0	0	(12)	
	CDH26.7	41	0	0	0	0	0	2	22	13	4	0	0	0	(13)	
(14) Wind	WSAvg	4.3	4.7	4.5	4.8	4.6	4.4	3.9	3.5	3.4	3.9	4.1	4.5	4.7	(14)	
(15) Precipitation	PrecAvg	1217	112	94	98	92	94	94	83	88	96	112	119	132	(15)	
	PrecMax	1579	263	176	191	178	218	202	232	210	178	215	216	259	(16)	
	PrecMin	848	32	29	45	14	26	20	9	9	23	29	40	64	(17)	
	PrecStd	173	40	35	34	35	44	42	47	49	39	47	39	47	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.4	8.2	11.8	18.2	24.3	26.4	28.9	28.2	26.2	21.3	17.2	12.4	(19)	
		MCWB	10.6	6.9	6.9	13.5	16.4	18.9	21.8	21.3	20.9	17.2	15.5	11.6	(20)	
	2%	DB	6.6	5.1	7.1	15.1	20.7	24.2	27.0	26.5	23.8	18.6	15.0	9.3	(21)	
		MCWB	6.0	4.1	5.6	11.8	15.0	17.9	20.9	20.3	19.2	15.7	13.8	8.6	(22)	
	5%	DB	4.1	2.7	5.0	12.2	18.0	22.5	25.8	25.2	22.1	16.9	12.9	6.8	(23)	
		MCWB	3.3	1.6	3.1	9.5	13.0	17.3	20.0	19.5	18.6	14.6	11.4	6.0	(24)	
	10%	DB	1.7	0.9	3.0	9.8	15.7	20.6	24.2	23.9	20.5	15.2	10.7	4.4	(25)	
		MCWB	0.9	-0.1	1.4	7.1	12.0	16.3	19.2	18.9	17.6	12.9	9.3	3.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.4	7.1	8.3	15.2	18.1	21.3	22.9	22.7	21.7	18.8	15.8	11.7	(27)	
		MCDB	11.3	7.7	10.1	17.6	22.1	24.6	27.0	26.4	24.8	20.3	17.0	12.5	(28)	
	2%	WB	6.1	4.3	5.8	12.5	16.0	19.7	22.0	21.7	20.2	16.3	14.0	8.7	(29)	
		MCDB	6.5	5.0	7.0	14.2	18.9	22.4	25.8	24.9	22.5	18.1	15.0	9.2	(30)	
	5%	WB	3.4	1.8	3.5	9.9	14.3	18.3	21.1	20.8	19.2	14.9	11.8	6.1	(31)	
		MCDB	4.0	2.5	4.7	11.7	17.0	21.0	24.3	23.9	21.4	16.6	12.6	6.7	(32)	
	10%	WB	1.0	0.1	1.4	7.4	12.4	16.7	20.2	19.8	18.0	13.2	9.3	3.8	(33)	
		MCDB	1.6	0.8	2.8	9.6	15.2	19.8	23.1	22.9	20.1	14.9	10.5	4.4	(34)	
(35) Mean Daily Temperature Range	MDBR		7.3	7.1	6.6	7.4	8.5	8.2	7.9	7.8	7.7	6.6	6.3	6.1	(35)	
	5% DB	MCDBR	10.8	9.3	9.0	11.5	12.7	11.0	10.2	9.6	9.3	8.5	9.4	9.4	(36)	
		MCWBR	11.0	8.8	7.9	8.6	8.0	6.5	5.5	5.0	6.1	6.7	9.0	9.7	(37)	
	5% WB	MCDBR	10.7	9.6	9.0	10.5	11.1	9.4	8.6	8.2	8.2	8.2	9.3	9.2	(38)	
		MCWBR	11.0	9.4	8.6	9.0	8.0	6.4	5.3	5.0	6.2	7.3	9.3	9.5	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.299	0.288	0.318	0.347	0.365	0.381	0.406	0.393	0.355	0.333	0.312	0.290	(40)	
	taud		2.339	2.304	2.320	2.399	2.417	2.388	2.354	2.388	2.490	2.526	2.506	2.464	(41)	
	Ebn at Noon		791	887	908	910	901	885	854	861	832	783	774	774	(42)	
	Edh at Noon		75	94	108	110	112	116	119	111	92	76	64	60	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.06	1.77	3.13	4.19	5.10	5.55	5.64	4.98	3.80	2.35	1.21	0.82	(44)	
	RadStd		0.07	0.24	0.30	0.38	0.46	0.42	0.38	0.31	0.33	0.24	0.15	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (41 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.35	N/A	N/A	N/A	N/A

Nomenclature: See separate page